SPHERE: MISSION CONTROL

STUDENT MANUAL, EXPERIMENTAL METHOD AND ANALYSIS GUIDE

Experimental objective: Construct an insulated $100\ \mathrm{mL}$ water capsule, expose it to an ice-water boundary ($0^\circ\mathrm{C}$) and record the temperature for $15\ \mathrm{minutes}$. Compare the measurements with Newton's law of cooling using residuals and MAE ($\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\Delta T_i|$).

WORKFLOW

RECOMMENDED SEQUENCE

To achieve maximum learning outcome and experimental precision, follow the **5-Stage Master Execution Flowchart** below. Complete all activities in sequence: review theory first, assemble hardware, collect lab data, complete your Task Sheet analysis, and test knowledge with Flash Cards.

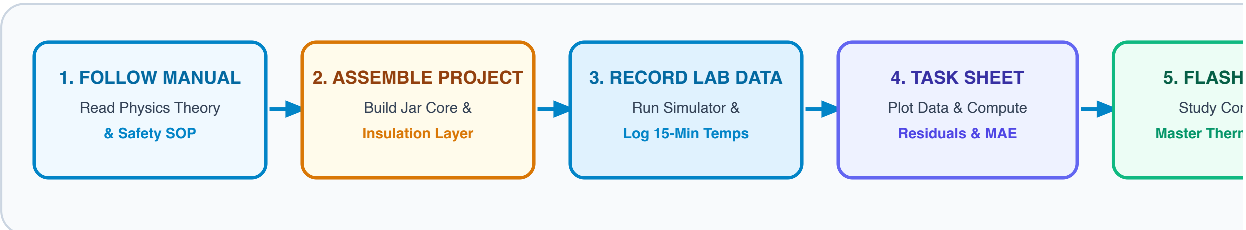


FIGURE 1: END-TO-END STUDENT LABORATORY & TASK COMPLETION FLOWCHART

1 FOLLOW MANUAL FIRST

Read Sections 1–4 of this manual. Review Newton's Law of Cooling, heat transfer mechanisms (conduction, convection, radiation), and experimental safety SOP before handling hot water.

2 ASSEMBLE PROJECT

Assemble the $100\ \mathrm{mL}$ core water container, seal the temperature probe, and apply your team's assigned insulation configuration (Bare Control, Cotton Fiber, Bubble Wrap, or Reflective Film).

3 RECORD LAB DATA

Run theoretical predictions on Mission Control Console (`index.html`). Lower capsule into $0^\circ\mathrm{C}$ ice bath and record temperatures every 60s for 15 mins in Mission Control and your paper log.

4 COMPLETE TASK SHEET

Open [Student Task Sheet]. Log readings, plot curves, compute temperature residuals ($|\Delta T|$) and MAE score, and answer analytical evaluation questions.

REVIEW FLASH CARDS

Open **SPHERE Flash Cards**. Test your understanding of thermal insulation concepts, Newton's law formulas, residual calculations, and aerospace applications.

1. ENGINEERING CONTEXT & OBJECTIVES

Spacecraft thermal control must account for radiative exchange, internal heat generation and highly variable external heat fluxes. In vacuum there is no external convection; thermal design therefore differs fundamentally from this water-bath experiment.

Aerospace multilayer insulation uses low-emissivity reflective layers separated by low-conductance spacers. The cotton, bubble wrap and reflective film used here form an educational multilayer assembly, not flight-qualified MLI.

Test the assigned insulation configuration under a common boundary condition. Compare observations with the analytical reference curve and report residuals, MAE, uncertainty sources and model limitations.

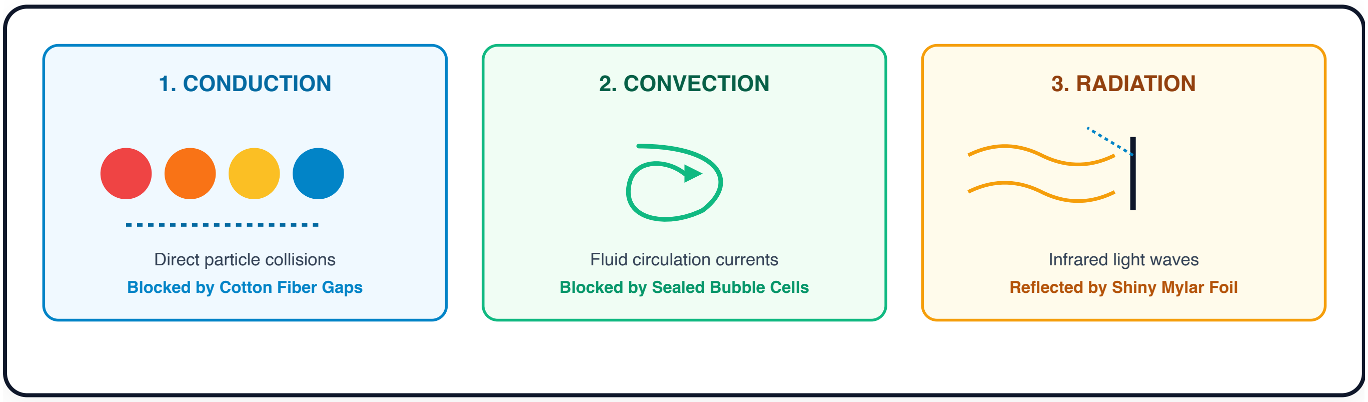


FIGURE 2: THREE MODES OF HEAT TRANSFER & THERMAL INSULATION BARRIERS

2. HARDWARE SETUP & SCHEMATIC

Assemble a sealed test capsule suitable for controlled immersion:

HARDWARE PART	DESCRIPTION	WHAT IT DOES IN OUR LAB
Core Jar	$100\ \mathrm{mL}$ wide-mouth glass jar	Holds our hot water core ($T_0 = 80.0°\mathrm{C}$), representing the astronaut's internal body heat.
Prepared Lid	Jar lid with a probe opening	Supports consistent probe placement; seal using the approved lab method and leak-test before use.
Digital Sensor	Waterproof digital thermometer	Measures internal core water temperature (T_{obs}, accuracy $\pm 0.5°\mathrm{C}$). Record the accuracy for your instrument.
Ice Bath Chamber	Ice-water slurry ($0.0°\mathrm{C}$)	Provides a repeatable low-temperature boundary sink ($T_{\mathrm{env}} = 0.0°\mathrm{C}$) for comparative testing.

Quick Assembly Steps:

1. **Feed the Sensor:** Slide the metal temperature probe through the prepared opening in the jar lid.
2. **Set the Depth:** Position and secure the lead so the probe remains at the required depth.
3. **Seal and Test:** Seal the probe opening using the approved lab method, close the lid, and complete a leak test before immersion.
4. **Center the Probe:** Make sure the metal sensor tip hangs right in the center of the liquid without touching the bottom or sides.

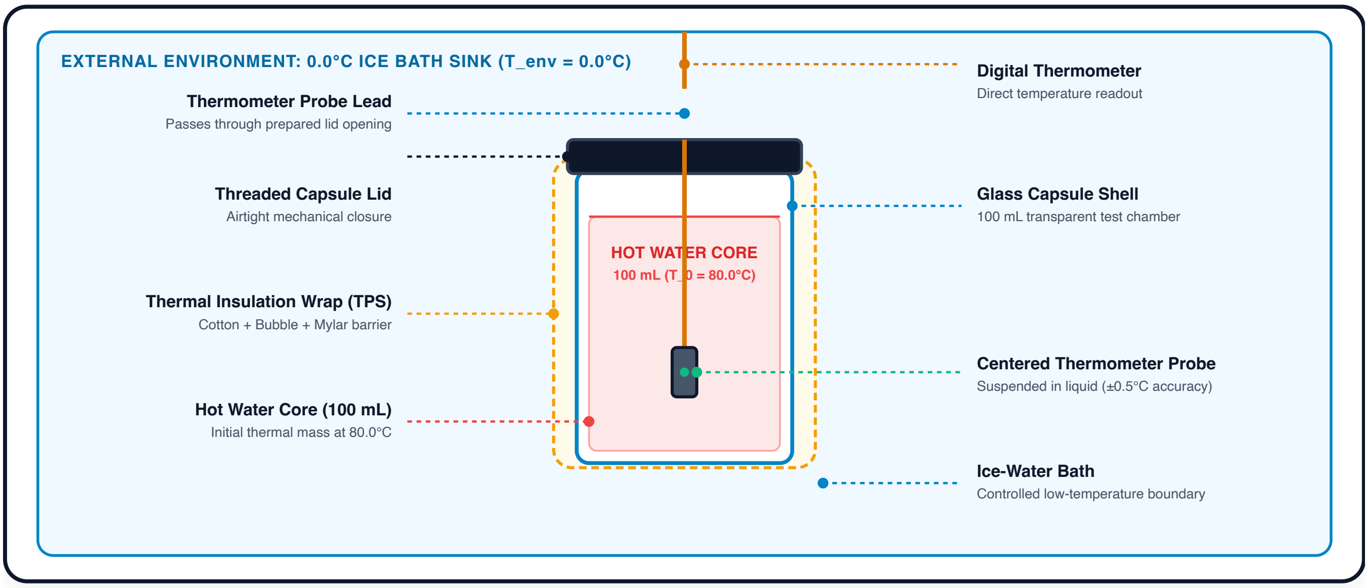


FIGURE 1: EXPERIMENTAL HARDWARE ASSEMBLY & SENSOR ALIGNMENT

3. THE 4 INSULATION CONFIGURATIONS

Each team tests one of four configurations under the same nominal boundary conditions ($T_0 = 80.0^\circ\mathrm{C}, T_{\mathrm{env}} = 0.0^\circ\mathrm{C}$):

A. BARE CONTROL $k \approx 0.150\ \mathrm{min}^{-1}$

Plain uninsulated jar. Shows how rapidly heat escapes through direct conduction and water currents.

B. BUBBLE WRAP $k \approx 0.040\ \mathrm{min}^{-1}$

Two even layers of bubble wrap. Sealed air cells reduce fluid motion and slow heat transfer.

C. MYLAR FOIL $k \approx 0.121\ \mathrm{min}^{-1}$

Shiny emergency blanket. Reflects radiant thermal infrared waves back toward the hot water.

D. MULTILAYER ASSEMBLY $k \approx 0.015\ \mathrm{min}^{-1}$

Combines cotton, bubble wrap and reflective film to increase effective thermal resistance.

Optional extension: Add a fifth cotton-only trial to isolate the contribution of the inner fibrous layer.

FIGURE 3: EDUCATIONAL MULTILAYER TEST ASSEMBLY

4. THE PHYSICS: NEWTON'S LAW OF COOLING

Heat always flows from warmer objects to cooler surroundings. Over 300 years ago, Sir Isaac Newton discovered that the rate at which an object cools is proportional to the temperature difference between the object and its environment:

NEWTON'S LAW OF COOLING (EXPONENTIAL DECAY MODEL)

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-k \cdot t}$$

Where:

$T(t)$ = Predicted core temperature ($^\circ\mathrm{C}$) at minute t

T_{env} = Ice bath temperature ($0.0^\circ\mathrm{C}$)

T_0 = Starting core temperature ($80.0^\circ\mathrm{C}$)

k = Cooling rate constant (min^{-1})

Interpretation of k: The fitted cooling constant k has units of min^{-1}. Under otherwise identical conditions, a smaller value corresponds to a slower approach to the bath temperature.

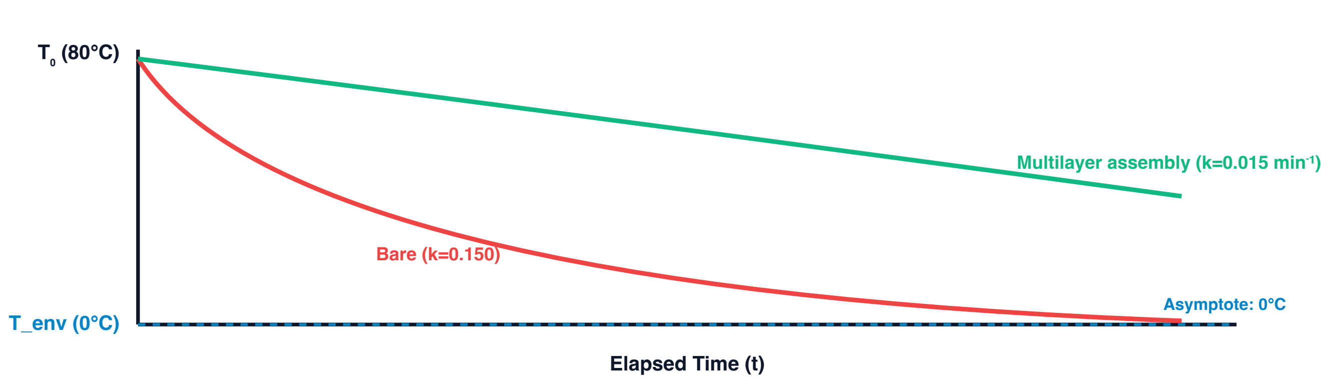
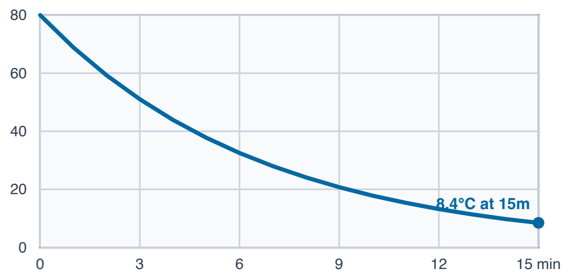


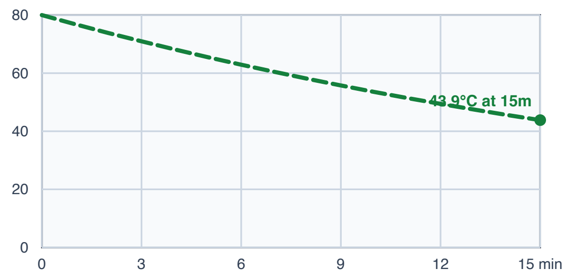
FIGURE 4: EXPONENTIAL DECAY PROFILES ACROSS HIGH VS LOW COOLING CONSTANTS (k)

INDIVIDUAL PREDICTION CURVES ($T_0 = 80.0°\text{C}, T_{\text{ENV}} = 0.0°\text{C}$)

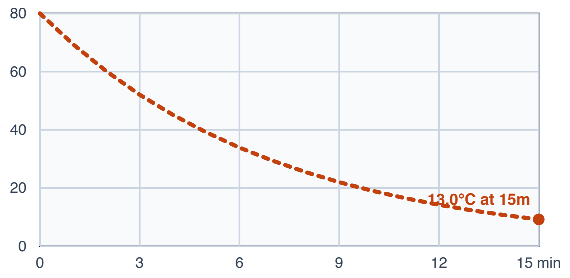
A. BARE CONTROL ($k \approx 0.150\ \text{min}^{-1}$)



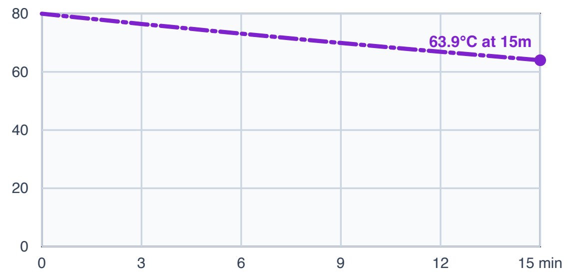
B. BUBBLE WRAP ($k \approx 0.040\ \text{min}^{-1}$)



C. MYLAR FOIL ($k \approx 0.121\ \text{min}^{-1}$)



D. MULTILAYER ASSEMBLY ($k \approx 0.015\ \text{min}^{-1}$)



5. STEP-BY-STEP LAB WORKFLOW

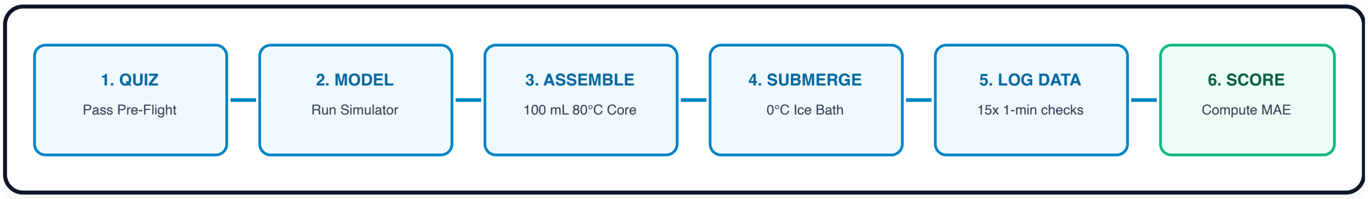


FIGURE 5: 6-STAGE EXPERIMENTAL PROTOCOL WORKFLOW

1. **Complete the concept check:** Review the heat-transfer questions before handling hot water.
2. **Generate Model Prediction:** Go to the **Simulator** tab, choose your team's assigned insulation configuration, verify starting conditions ($T_0 = 80.0°\mathrm{C}$, $T_{\mathrm{env}} = 0.0°\mathrm{C}$), and click **RUN SIMULATION PREDICTION**.
3. **Fill the Water Core:** Carefully measure $100\ \mathrm{mL}$ of hot water at $80.0°\mathrm{C}$, pour it into your jar, screw the cap on tightly, and verify the seal is leak-free.
4. **Submerge and start timing:** Lower the capsule into the measured ice-water bath $(0.0°\mathrm{C})$ and immediately start the timer in the **Measurement Log** tab.
5. **Log Temperature Every Minute:** Every $60\ \mathrm{seconds}$ for $15\ \mathrm{minutes}$, write down the observed temperature in your Task Sheet and enter it into Mission Control.
6. **Evaluate:** Plot the measured and predicted curves, calculate residuals ($|\Delta T| = |T_{\mathrm{obs}} - T_{\mathrm{model}}|$) and MAE $(\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\Delta T_i|)$, and discuss the principal sources of uncertainty.